

# Ankit Shukla

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## SUMMARY

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Software engineer with 15 years of experience building C/C++ systems software across Linux and distributed infrastructure. Strong background in Linux systems programming, concurrency, multithreading, IPC, performance debugging, and production support. Currently developing cloud-native control-plane software for VMs, disks, and containerized workloads using C++, Go, Python, gRPC, REST, and Kubernetes.

## EDUCATION

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Uttar Pradesh Technical University (UPTU) — B.Tech, 2010

Pune University — Diploma in System Software, 2011

## EXPERIENCE

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### Software Engineer 4 — Resource Manager

Dec 2024 – Present

COHESITY INC., Pune

- Design and implement core components of a cloud-native control-plane resource manager.
- Develop services for provisioning and tracking Linux/Windows VM fleets, disks, and containerized workloads.
- Build services using C++, Go, Python, gRPC, REST, and Kubernetes.
- Own architecture, implementation, testing, production support, and root-cause analysis.
- Integrated cloud networking and compute across AWS and Azure — EC2, Azure VMs, VNets, subnets, Internet Gateway (IGW), and NAT Gateway — to provision multi-cloud VM and container infrastructure.
- Implemented secure service communication using OAuth2, JWT, HTTPS, HTTP/2, TCP/IP, and SSH, fronted by load balancers, reverse proxies, API gateways, and rate limiters.
- Built synchronous and event-driven service integrations — REST/RESTful, SOAP, and WebSocket APIs alongside Kafka and RabbitMQ messaging — spanning monolithic, microservices, and event-based architecture styles.
- Managed cloud identity and infrastructure-as-code using AWS IAM and Azure ARM templates.

### Principal Software Engineer

Jan 2023 – Dec 2024

VERITAS TECHNOLOGIES, Pune

- Developed C++ licensing and entitlement software used across multiple product platforms.
- Ported and validated software across Windows, Linux, AIX, and Solaris.
- Managed SDK delivery, build infrastructure, OpenSSL/POCO dependencies, CVE remediation, and SBOM generation.

### Senior Software Engineer / Technical Lead

Mar 2019 – Jan 2023

SIEMENS MOBILITY, Pune

- Led six engineers and owned end-to-end development of C/C++ railway control software.
- Drove requirements, architecture, implementation, testing, release, code reviews, and mentoring.
- Built Python automation improving engineering environment and test management.

### Specialist — Automotive Embedded Systems

Jun 2015 – Mar 2019

TATA ELXSI, Pune

- Developed C/C++ software on QNX and Windows.
- Worked on system integration, CAN communication, debugging, and root-cause analysis.

### Senior Software Engineer — Graphics Driver Development

Aug 2014 – Jun 2015

INFOSYS, Hyderabad

- Developed and debugged Linux systems software using Mesa, LLVM, libdrm, and DRM/KMS.
- Investigated GPU command submission and performance-critical rendering paths.

### Software Engineer

Aug 2011 – Aug 2014

ROCKWELL COLLINS, Hyderabad

- Developed C/C++ systems software on LynxOS.
- Worked with synchronization, interrupts, MMIO, hardware registers, and buffer queues.
- Managed system requirements using DOORS, and worked with ARINC653, ARINC429, and DO-178B avionics software standards.

## TECHNICAL SKILLS

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**Languages:** C++, C, Go, Python

**Distributed Systems:** Microservices, Monolithic, Event-Driven Architecture, Control Plane, gRPC, REST/RESTful, SOAP, WebSocket

**Messaging:** Kafka, RabbitMQ

**Cloud:** Kubernetes, Docker, AWS (EC2, IAM), Azure (VMs, VNets, ARM Templates), SQL, NoSQL

**Networking & Security:** TCP/IP, HTTPS, HTTP/2, SSH, OAuth2, JWT, Load Balancer, Reverse Proxy, API Gateway, Rate Limiter, IGW, NAT

**Systems:** Linux, POSIX, IPC, Multithreading, Synchronization

**Avionics/Requirements:** DOORS (Requirements Management), ARINC653, ARINC429, DO-178B

**Tools:** GoogleTest/GoogleMock, Ginkgo, CMake, Bazel, Jenkins, Git, JFrog

**Security:** OpenSSL, POCO, Black Duck, CVE Analysis, SBOM